What minimum number of ground stations is needed? Give the position equations.

(II) Stations D_1, D_2 lie on one meridian at θ N and θ S (unit: (°))
Both receive a satellite signal τ after emission. (i) Find its maximum altitude H above ground; (ii) For small latitude errors $\triangle\theta$ at D_1, D_2, find the error in H (iii) For a small error $\triangle\tau$ in time τ, find the error in H.

2. In 1 (II), take $\theta = 45°$, $\tau = 0.100$ s. (i) Find the maximum altitude H in km. (ii) If $\theta = \pm 1.0''$, how large is the error in H? (iii) If $\triangle\tau = \pm 0.010\ \mu$s, what error in H? Earth is a sphere of radius $R = 6.378 \times 10^3$ km; $c = 2.998 \times 10^8$ m/s, and ground gravitational acceleration $g = 9.81$ m/s^2.

3. Now use reference-satellite motion to track an object. Neglect relativity. Each continuously emitted radio signal states its emission time and the satellite's position then. A receiver aboard the object, with no clock, obtains these data. To infer its motion state, its position and time on receiving a signal, what minimum number of different satellites must it receive simultaneously? Write the equations for the object's position and time at reception.

4. Special relativity slows moving clocks; general relativity slows clocks in a gravitational field. Include both effects. Satellite and station clocks have aligned zero readings. Orbit height $h = 2.000 \times 10^4$ km; radius R, light speed c, and ground acceleration g are as given in part 2.

(I) By special relativity, estimate the reading difference after 24 h on Earth: ground minus satellite. Use uniform rectilinear time dilation for circular motion.

(II) The GR slowing factor is $(1-2\phi/c^2)^{1/2}$, where ϕ is the magnitude of gravitational potential: potential energy per unit mass, zero at infinity. After 24 on the ground clock, find the satellite-minus-ground clock reading.

3.

A refrigerator sends heat from a cold reservoir, plus energy from external work, to a hot reservoir, cooling the cold one and warming the hot one. If it works between absolute temperatures T_1 (hot) and T_2 (cold), taking heat